PROJECT REPORT ON OBJECT DETECTOR

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1. Introduction

Visually impaired people often have a really hard time getting around inside places they do not know well. Like spotting everyday stuff such as a bottle or a phone, or even something simple like a chair. It just makes navigation tricky.

This project aims to help with that. By building something called a Real-Time Intelligent Agent. It acts like digital eyes for them. I think that is a cool way to put it.

The system uses computer vision and supervised learning to handle video from a webcam right away. It looks at the feed, picks out objects, and gives feedback on the spot. So a regular laptop camera could become this helpful tool. Not sure if it covers everything perfectly, but it seems promising for indoor spots. That part might need more tweaking, though.

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2. Problem Statement

Traditional navigation aids (like white canes) detect physical obstacles but cannot identify what those obstacles are. There is a need for a low-latency, portable system that can:

1. Perceive the environment in real-time.
2. Classify objects with high confidence.
3. Communicate findings through natural language to the user.

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3. Methodology & Core Concepts

This project is built upon the four pillars of AI as taught in the **VITyarthi** curriculum:

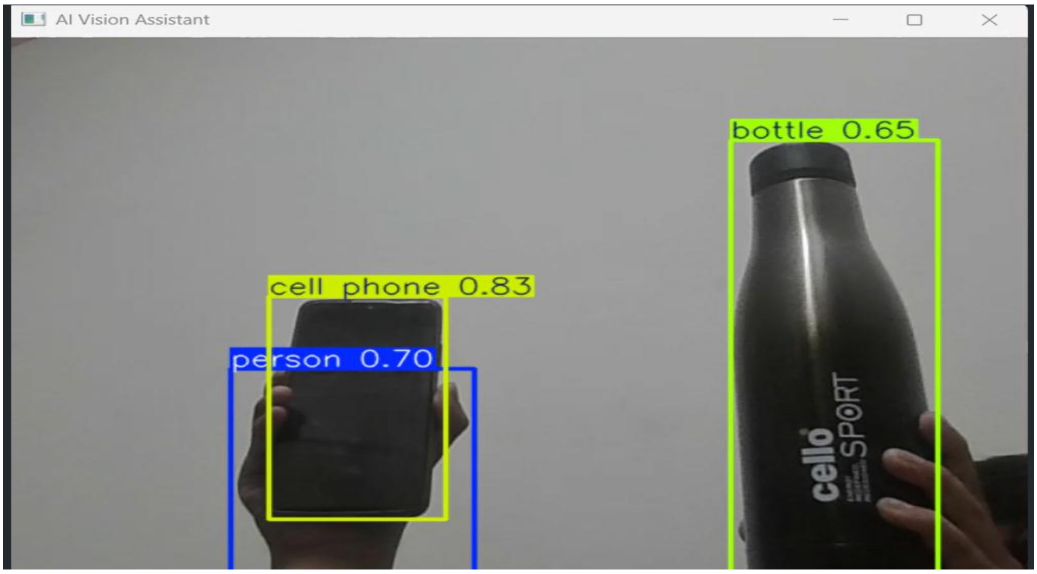
- **Intelligent Agent Framework:** The system follows the "Sensors-Agent-Actuators" model. The webcam (Sensor) sends data to the Python script (Agent), which triggers the Voice Engine (Actuator).

- **Object Detection (YOLOv8):** We utilized **Transfer Learning** by deploying the YOLOv8 (You Only Look Once) Nano model. This model uses a Deep Convolutional Neural Network (CNN) to perform Single Shot detection, making it fast enough for 1st-year hardware.
- **Confidence Thresholding:** To prevent False Positives (e.g., misidentifying fingers as toothbrushes), we implemented a confidence filter.
- **Text-to-Speech (TTS):** The **pyttsx3** library was used to convert the classification labels into speech, providing an accessible interface.

4. System Implementation

The implementation was carried out in **VS Code** using Python 3.x. The workflow is as follows:

1. **Input:** Stream live video using cv2.VideoCapture.
2. **Processing:** Each frame is resized and passed through the YOLOv8 neural network.
3. **Inference:** The model predicts bounding boxes and class probabilities.
4. **Filter:** The script filters for Useful Objects (Bottle, Phone, Chair, toothbrush, mouse, etc.).
5. **Output:** Visual bounding boxes are displayed on screen, and the highest-confidence object is announced via the audio engine.
    - Example output :-



5. Results & Discussion

During testing, the system successfully identified common household objects with an average confidence score of **75% to 85%**.

- **Challenges:** We observed "spatial ambiguity" where the model occasionally misidentified small, thin objects. This was mitigated by increasing the confidence threshold and narrowing the object interest list.
- **Performance:** The Nano version of the model allowed the script to run at smooth frame rates on a standard student laptop without the need for a dedicated GPU.

6. Conclusion

This **AI object detector** thing really shows how some basic AI ideas, like **PEAS** and **supervised classification**, plus those **rational agents**, get used to fix actual problems with accessibility. It feels like a good example, even if I'm not totally sure about every detail yet.

The prototype right now runs on a laptop, which is okay but not super handy. I think future versions might work on phones or even glasses you wear; something like with a Raspberry Pi setup. That would make it portable for people who can't see well, you know, actually helpful on the go.

It seems this project could build into more stuff later, like deeper learning techniques or how humans interact with computers in HCI. Some parts might get messy there, but it's a start anyway.